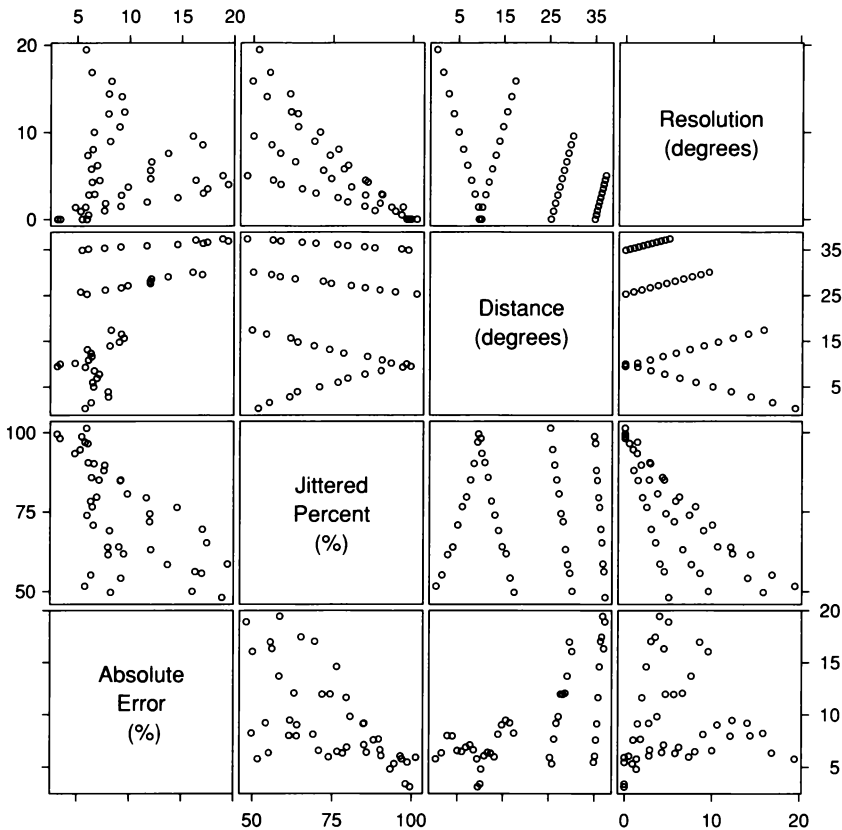


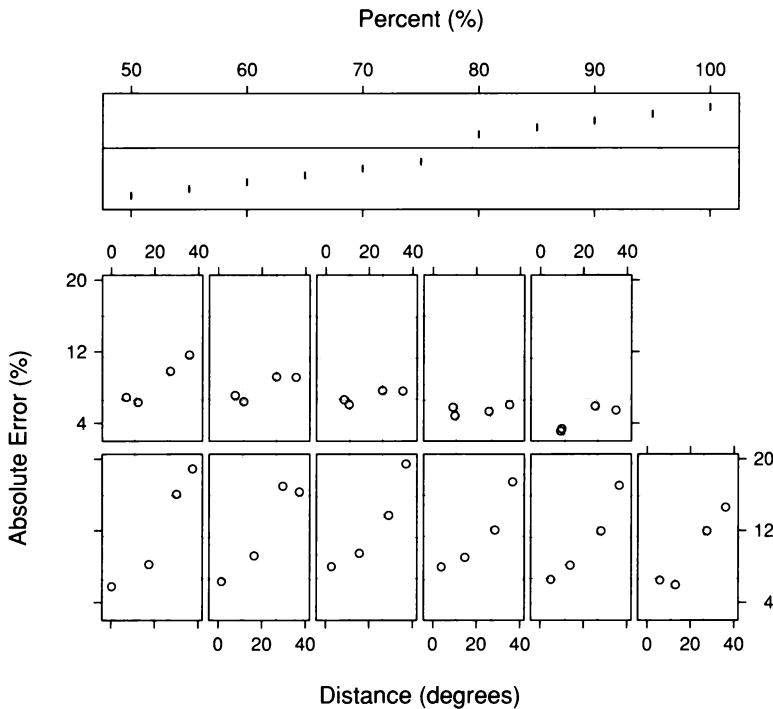
Exploring the Data

For the 44 line-segment pairs in the experiment, there are 44 values of a , d , p , and r . Figure 4.35 is a scatterplot matrix of the data; the values of p have been jittered to cure a problem of exact overlap. The (3,2) panel shows the 44 combinations of p and d that were used in the experiment. There are 11 distinct values of p and each one is repeated four times, although this cannot be appreciated from the figure because of the jittering. The values of d range from 0° to 37° .



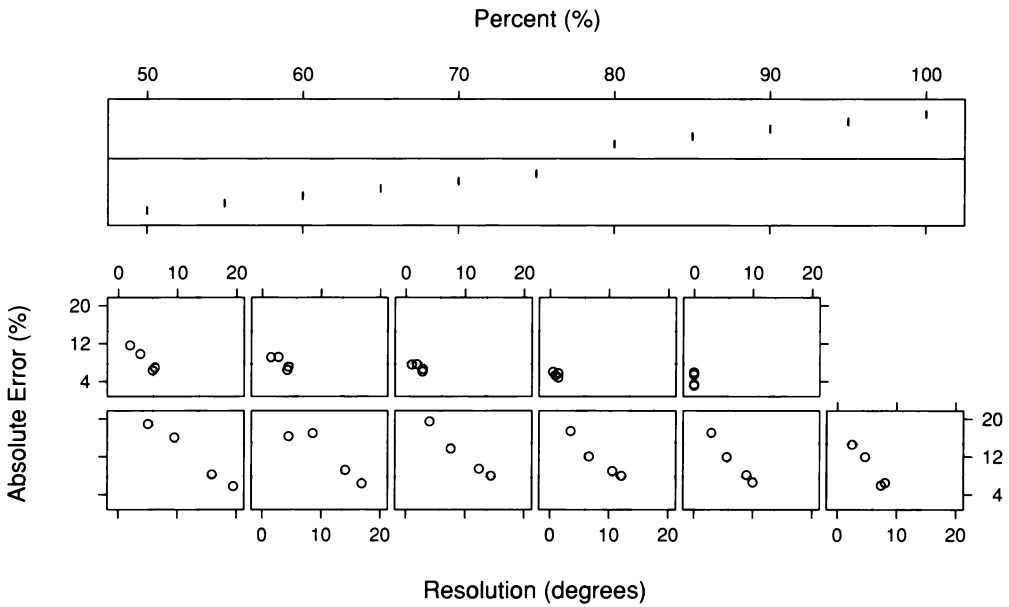
4.35 A scatterplot matrix displays the data from the experiment on judging slopes.

Figure 4.36 is a coplot of a against d given p . Because there are so few points on each dependence panel, loess curves have not been added. The conditioning values of p are the 11 distinct values of the measurements of p . Most panels have patterns with positive slopes, and for the smaller values of p , the patterns are slightly curved. Thus the hypothesis of the experiment — an increase of a as d increases — appears to be true. As p gets larger, the slope of the underlying pattern approaches the horizontal. (Note how we are making slope judgments in studying these data.) This means there is an interaction between p and d ; the effect on a of increasing d is greater when p is low than when it is high.

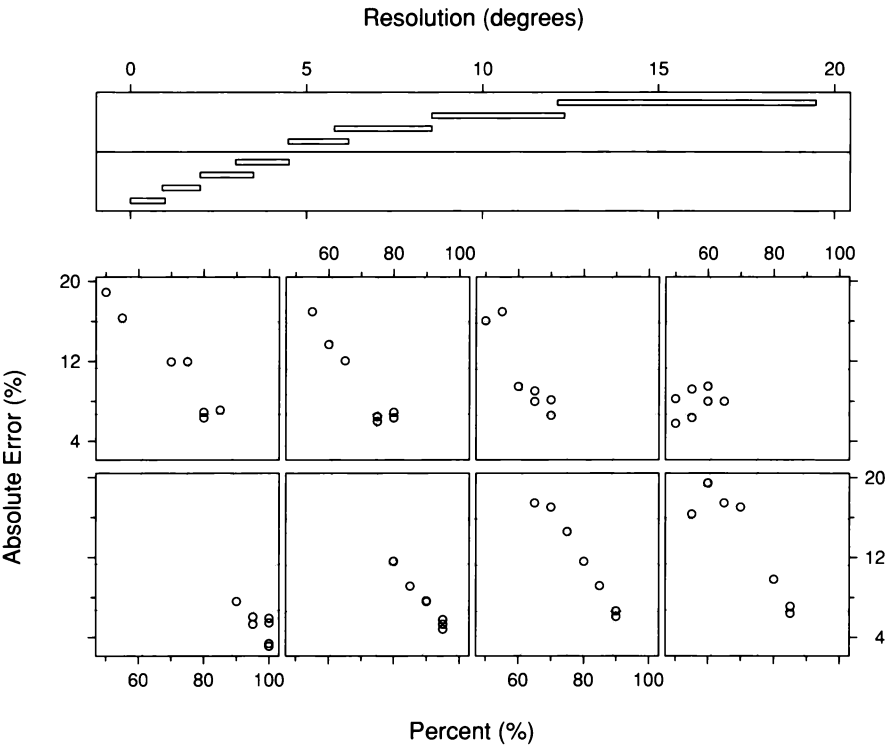


4.36 A coplot graphs absolute error against distance from 45° given percent. The aspect ratio has been chosen to bank the line segments between successive observations (not drawn) to 45° .

Figure 4.37 graphs a against r given p . The patterns are linear with no apparent curvature, and the slopes of the patterns, which are negative, do not appear to change, which means there is no interaction between p and r in explaining a . To probe the linearity and lack of interaction further, the coplot in Figure 4.38 graphs a against p given r . The equal-count method has been used to select the conditioning intervals; the number of intervals is 8 and the fraction of overlap is $1/4$. Again, the patterns appear linear and there is no interaction, confirming the effect seen in Figure 4.37.



4.37 A coplot graphs absolute error against resolution given percent. The line segments between successive observations (not drawn) are banked to 45° .



4.38 A coplot graphs absolute error against percent given resolution. The underlying patterns on the panels are banked by an aspect ratio of one.

As discussed earlier, the orientations of the line segments in the experiment are determined by p and d . They are also determined by p and r ; in other words, we can fit a to p and d , or to p and r . The coplots suggest that we should use r because the dependence has a simpler structure: a function linear in p and r . By contrast, an equation relating a to p and d would be more complicated, in part, because there is both curvature and an interaction between p and d . Once we have determined the dependence of a on p and r , we can use the equation that relates r to p and d to determine the dependence of a on p and d , and thus understand the interesting factor, d .

Fitting

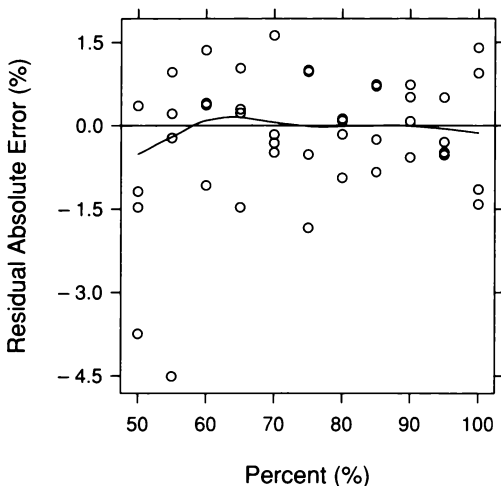
The coplot of a against r given p in Figure 4.37 shows that the residual variation about the underlying pattern is small. But as in previous data sets in this chapter, there appear to be a few observations that deviate by substantially more than the others, which suggests outliers among the residuals. Thus we will use bisquare to fit a linear function of p and r to a . The resulting fit is

$$a = 53 - 1.2r - 0.48p .$$

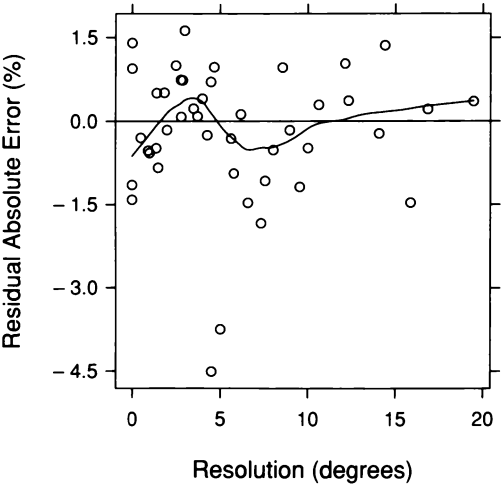
For this fit, an increase of 1% in p results in a decrease of 0.48% in a , and an increase of 1° in r results in a decrease of 1.2%.

Graphing Residuals

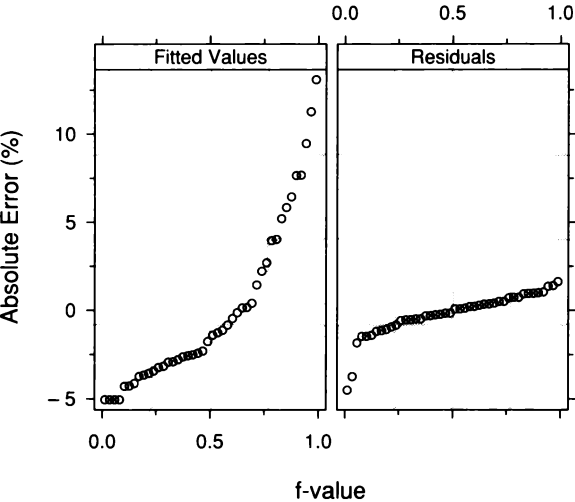
Figures 4.39 and 4.40 are residual dependence plots that graph the residuals against the factors. Figure 4.40 shows there is lack of fit, a convincing undulation that produces nearly all positive residuals for values of r between about 3° and 4° , and a preponderance of negative residuals between 5° and 10° . The graphs also reveal outliers for two observations with small values of p and values of r near 5° . Figure 4.41 is an r - f spread plot. Despite the lack of fit, the fitted function explains much of the variation in the absolute errors.



4.39 A residual dependence plot graphs residual absolute error against percent. The parameters of the robust loess curve on the plot are $\alpha = 1/2$ and $\lambda = 1$.



4.40 A residual dependence plot graphs residual absolute error against resolution. The parameters of the robust loess curve on the plot are $\alpha = 1/2$ and $\lambda = 1$.



4.41 An r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the bisquare fit to absolute error.

We will ignore the lack of fit in this case. The r-f spread plot has shown that the residual variation is exceedingly small, so any resulting improvement would not have an appreciable effect on the fitted surface. It is likely that we would need loess to remove the lack of fit; thus the cost of a negligible change in the surface would be the loss of the extreme simplicity of the linear fit. For this scientific application, the nonlinear surface is not worth the cost.

Visualizing the Dependence on Distance

The response, a , has been fitted by a function of p and r ,

$$a = 53 - 1.2r - 0.48p .$$

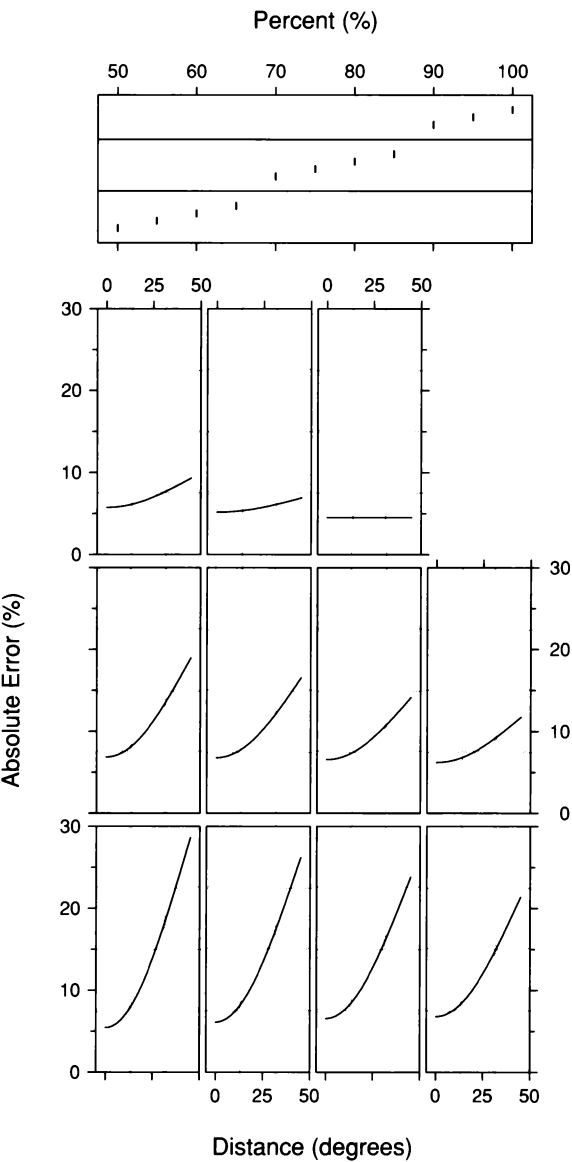
But it is d , the distance from 45° , that is of chief interest. However, r is related to p and d by

$$r = \arcsin \left[\frac{100 - p}{100 + p} \cos (2d) \right] .$$

Substituting for r in the fitted equation, we have

$$a = \hat{g}(d, p) = 53 - 1.2 \arcsin \left[\frac{100 - p}{100 + p} \cos (2d) \right] - 0.48p .$$

Figure 4.42 graphs this fitted function by a coplot of a against d given p . For all values of p , a does not change by much as d increases from 0° to about 10° . In other words, small deviations from 45° banking do not appreciably degrade the judgment of orientation. But once d gets above 10° , the error increases rapidly for some percents, particularly those close to 50%. However, for percents close to 100%, the absolute error is very small and d has little influence. Percents near 100% are judged very accurately because it is easy for our visual system to recognize that the segments lie on nearly parallel lines.



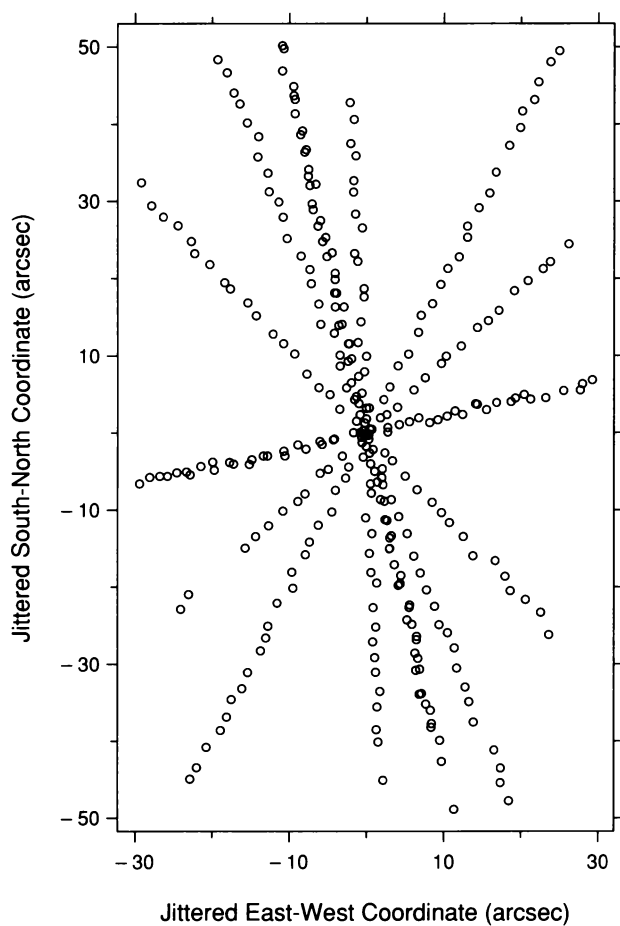
4.42 The surface fitted to absolute error by bisquare is graphed against distance given percent. The curves are banked to 45°.

4.6 *Level Plots of Data*

NGC 7531 is a spiral galaxy in the Southern Hemisphere. When looked at from the earth, the galaxy fills a very small area on the celestial sphere. If the only motion of NGC 7531 relative to the earth were the rapid recession due to the big bang, then over the entire region, the velocity relative to the earth would be constant and equal to about 1600 km/sec. But the actual motion is complex. The galaxy appears to be spinning, and there are other motions that are not well understood. The velocity at different points of the galaxy varies by more than 350 km/sec.

Figure 4.43 shows locations where 323 measurements were made of the galaxy velocity [15]. The values have been jittered to reduce overlap. The two scales, whose units are arc seconds, are east-west and south-north positions, which form a coordinate system for the celestial sphere based on the earth's standard coordinate system. Note that east and west are reversed because we are looking at the celestial sphere from the inside. The measurement locations lie along one of seven lines, or slits, that intersect in the center of the measurement region; of course, this does not appear to be exactly the case in Figure 4.43 because the points are jittered.

The goal in analyzing the galaxy data is to determine how the velocity measurements vary over the measurement region; thus velocity is a response and the two coordinate variables are factors. We want to determine regions in the plane where velocity is low, regions where it is high, and regions where it is approximately equal to any one of the values between the two extremes of 1409 km/sec and 1775 km/sec. There is no compelling need to study the data by conditioning on one coordinate and graphing against the other. This would endow the coordinates with an importance they do not have. The geocentric coordinate system imposed on the sky by man is clearly of no importance to the energy arriving now from the galaxy, which began its transit about 50 million years ago. Any other set of perpendicular axes could equally well serve as a coordinate system.

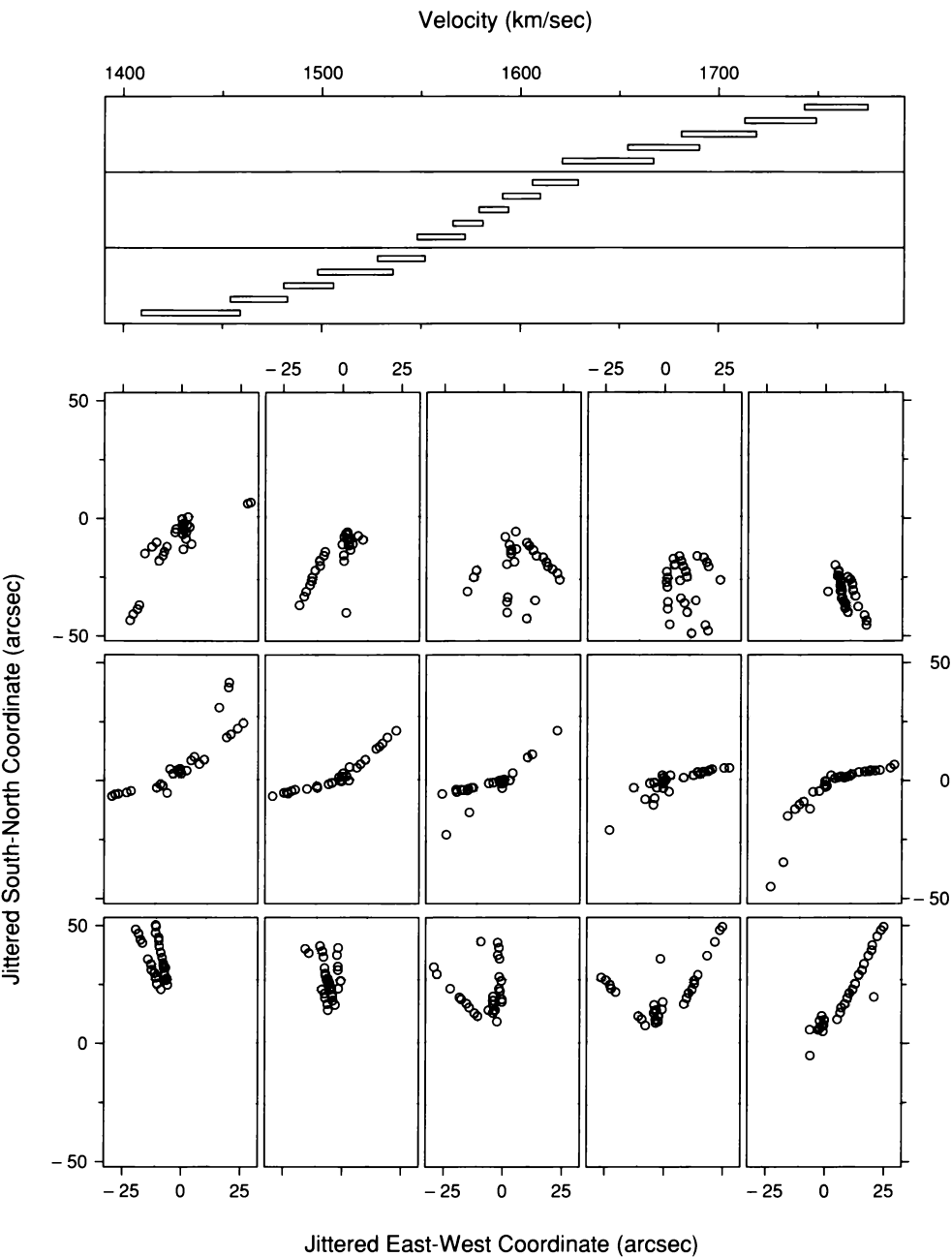


4.43 The plotting symbols show the locations of 323 measurements of the velocity of NGC 7531 relative to the earth. The aspect ratio has been chosen to make the number of data units per cm the same on both scales.

Level Regions

Figure 4.44 is a *level plot*, a visualization that shows regions with different levels of velocity. The format is that of a coplot, but instead of conditioning on a factor, as in a coplot, the conditioning is on the response. Velocity is broken up into different *levels*, the intervals shown in the *level panel* at the top of the figure. The intervals have been chosen using the equal-count method with the fraction of overlap equal to $1/4$. The 5×3 array of panels below shows *level regions*; each region is made up of those measurement locations where velocity is within one of the levels. The velocity levels increase as we move in graphical order through the level-region panels.

The (1,1) panel of Figure 4.44 shows that the region for the lowest velocity level lies in the northeast. As the levels increase, the regions move to the south and west, first as long narrow regions that are convex upward, then nearly straight regions, and then convex downward regions. Finally, the region for the highest level lies in the southwest.



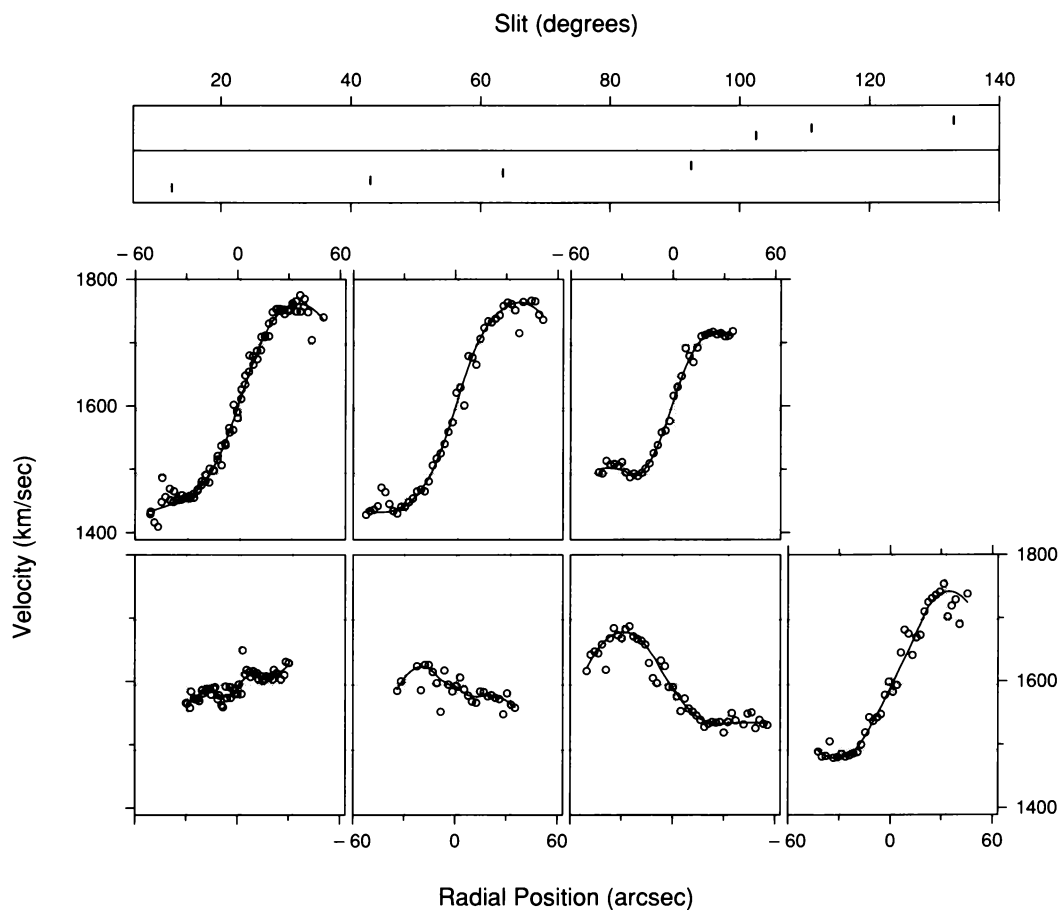
4.44 A level plot displays level regions of galaxy velocity. The aspect ratio of the level-region panels has been chosen to make the number of data units per cm the same on both scales.

4.7 *Improvisation*

One aspect of the structure of the galaxy data provides a special opportunity for further visualization. The measurements, as we have seen, lie along seven slits that intersect at the origin; for each slit separately, we can graph velocity against position along the slit. This, of course, is not a general method, but an improvisation, a visualization that is possible because of the special structure of this trivariate data set. Data analysis is difficult. Serendipity should be exploited.

Each slit can be described by the angle of a clockwise rotation to the horizontal; the seven angles are 12.5° , 43° , 63.5° , 92.5° , 102.5° , 111° , and 133° . Each location of a velocity measurement has a slit angle, the angle of the slit on which it lies. And each location has a radial position along the slit, which we will take to be distance of the location from the origin, multiplied by -1 if the east-west coordinate is negative and by 1 if it is nonnegative.

Figure 4.45 graphs velocity against radial position for each slit; this is achieved by a coplot in which slit angle is the conditioning variable. The underlying patterns of the seven sets of data in Figure 4.45 are varied and highly nonlinear. The overall amount of deviation of the data from the patterns is not large, but a number of observations deviate by considerably more than the others, which suggests leptokurtosis.

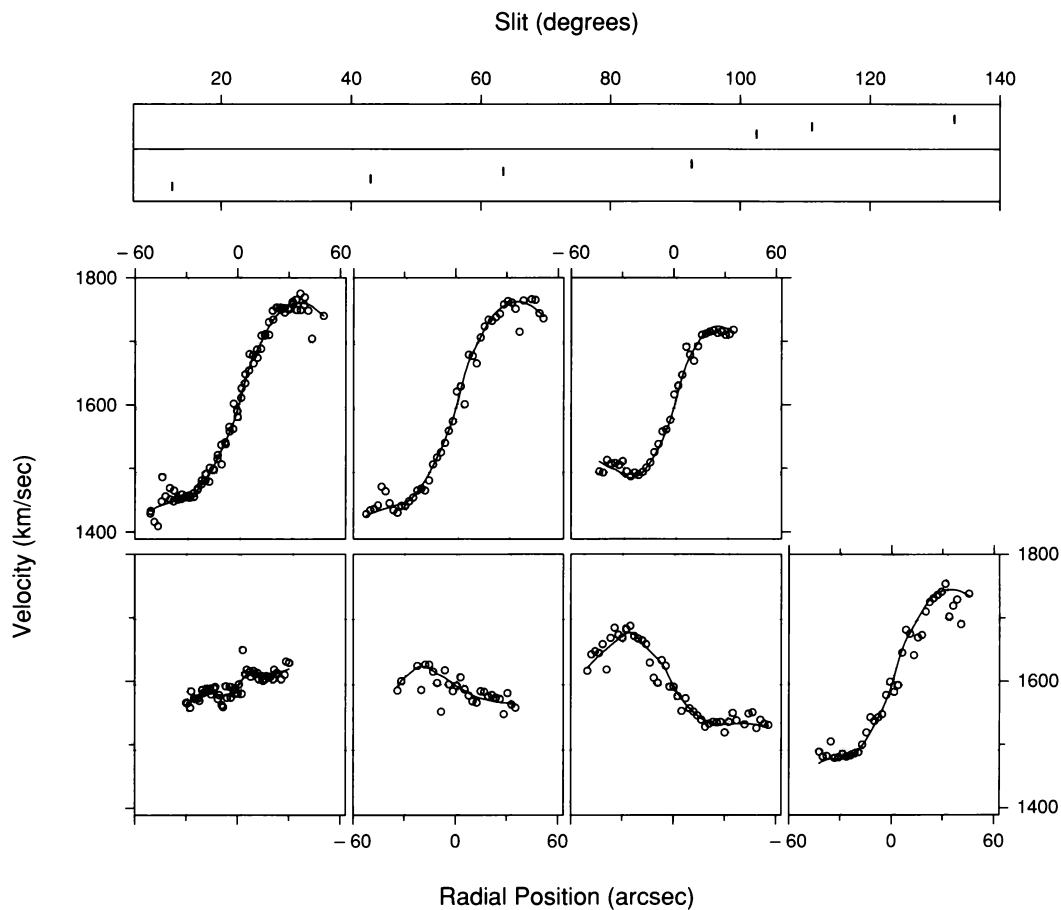


4.45 A coplot graphs galaxy velocity against radial position given slit angle. The parameters of the robust loess curves are $\alpha = 1/2$ and $\lambda = 2$. The curves are banded to 45° .

Fitting and Graphing Residuals

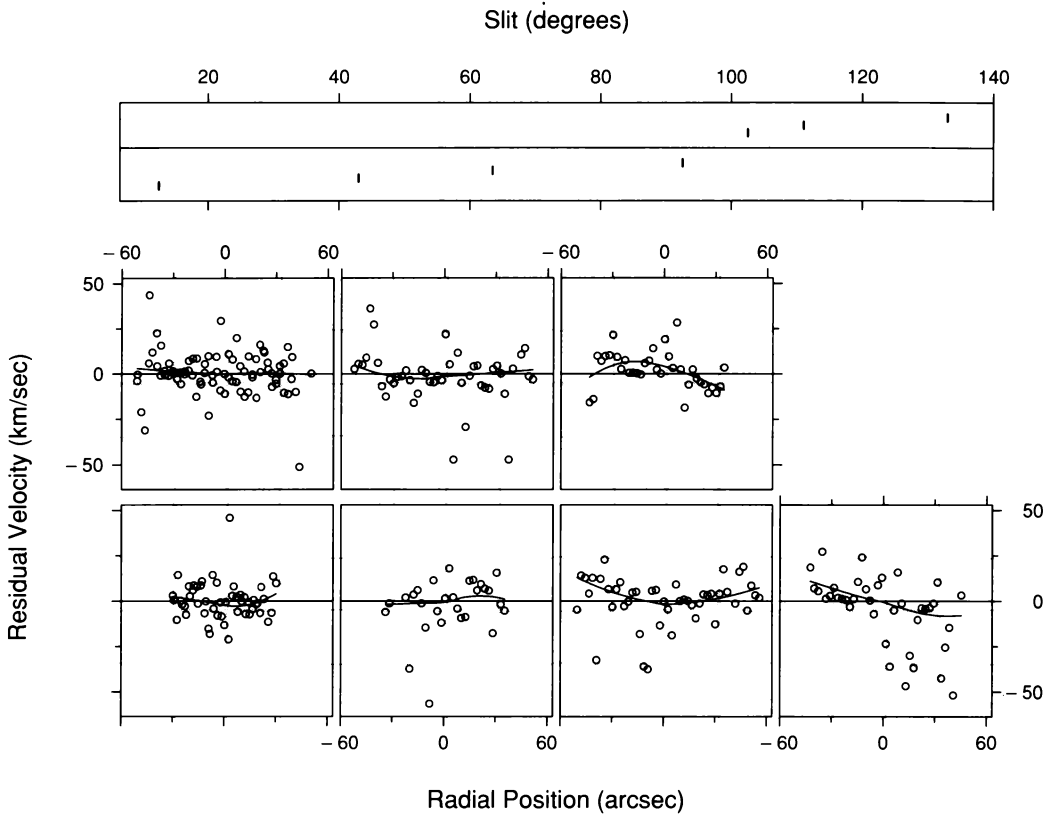
The next step in analyzing the galaxy data is to fit a surface. The level plot in Figure 4.44 and the coplot in Figure 4.45 show a complicated pattern with substantial curvature. We cannot expect a parametric surface to adequately track such a pattern. Loess fitting is needed. Because of the substantial curvature, λ must be two. Extensive experimentation led to a choice of $1/4$ for α ; this value struck a reasonable balance between lack and surplus of fit. Graphs of the fit and the residuals, to be shown shortly, will demonstrate this. As suggested by the coplot in Figure 4.45, the residual variation is leptokurtic, so bisquare is needed. Finally, to preserve the actual distance in the coordinate system of the celestial sphere, the loess fitting omits the standardization of the measurements of the two factors by their 10% trimmed standard deviations.

The slit structure of the galaxy data can also be used to see how well the loess surface fits the data. Figure 4.46 is a coplot of the data; the curves are the fitted surface evaluated along the slits. The fit is excellent and does almost as well as the robust loess curves fitted separately to the data of each panel in Figure 4.45.



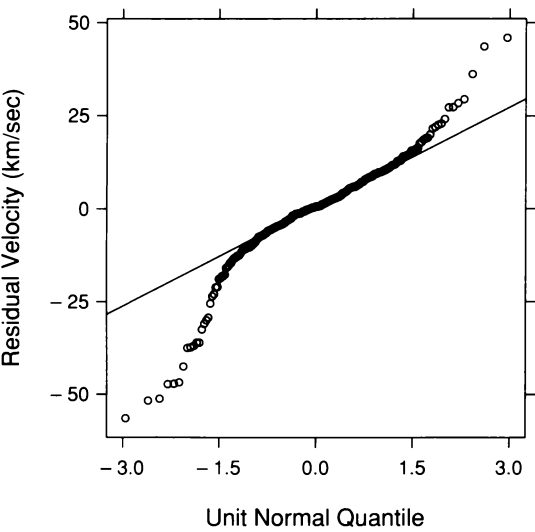
4.46 The coplot of the data graphs velocity against radial position given slit angle. The curves are the robust loess surface evaluated along the slits. The curves are banked by taking the aspect ratio of the dependence panels to be the same as that of Figure 4.45.

Figure 4.47 is a coplot of the residuals. There is lack of fit, a remaining dependence of the location of the residuals on radial position. In panel (4,1), there is a slight downward slope, and in panel (3,2), most of the residuals for radial positions less than zero are positive and most for radial positions greater than zero are negative. While we have detected lack of fit, its magnitude is quite small; we will endure it rather than endure the decrease in the smoothness of the surface that would result from curing it. Figure 4.47 also suggests there is nonhomogeneous spread; for example, in panel (4,1), there is a sudden change in the spread of the residuals at the origin.

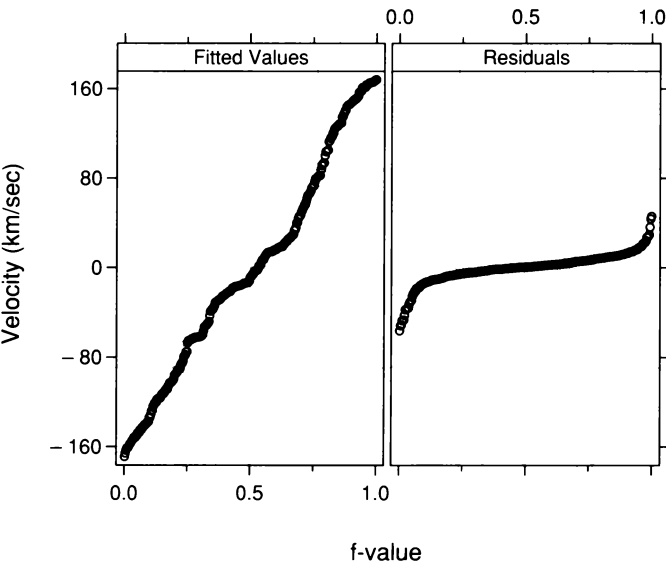


4.47 The velocity residuals are graphed against radial position given slit angle. The parameters of the robust loess curves are $\alpha = 1$ and $\lambda = 2$.

Figure 4.48, a normal quantile plot of the residuals, reveals significant leptokurtosis. Figure 4.49 is an r-f spread plot; the surface explains a major portion of the variation in the velocities. In the next three sections we will show further visualizations of the fit.



4.48 A normal q-q plot compares the distribution of the velocity residuals with the normal distribution.



4.49 An r-f spread plot compares the spread of the residuals and the fitted values minus their mean for the robust loess fit to the galaxy velocities.

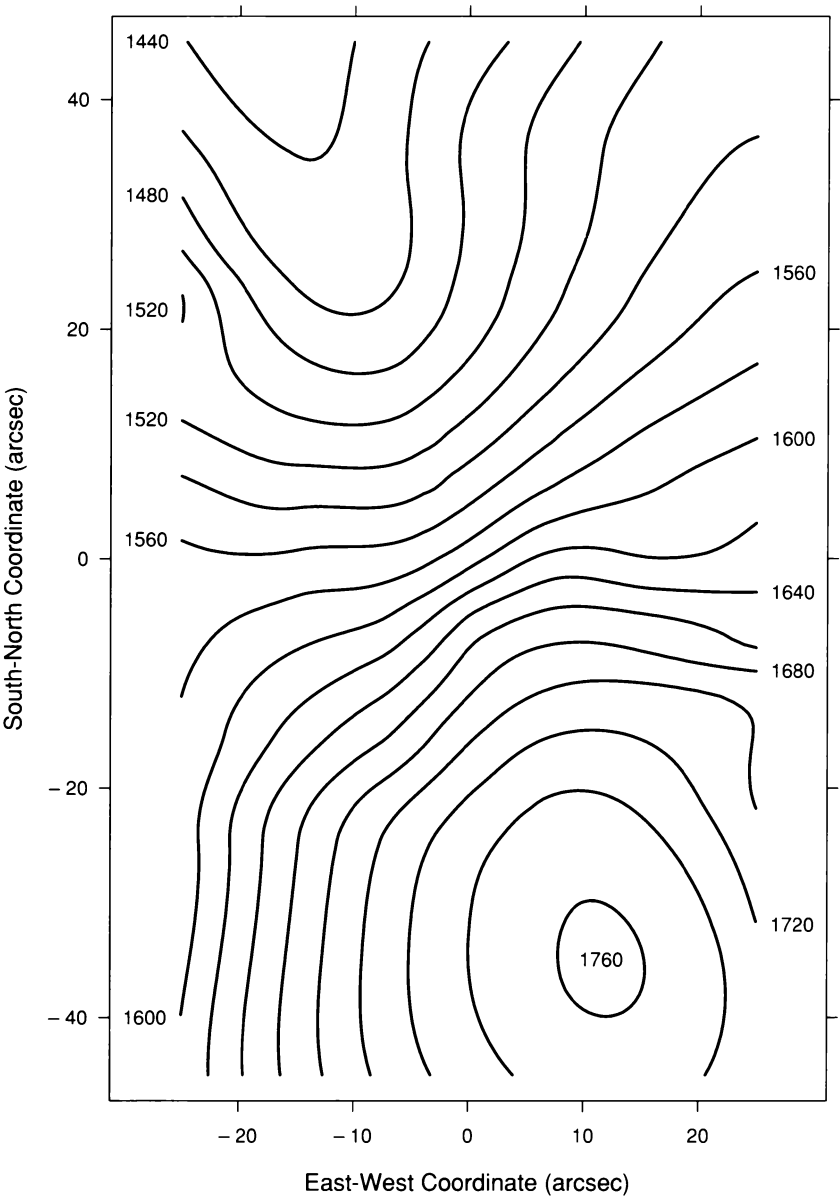
4.8 Contour Plots of Surfaces

For decades, the *contour plot* has been a workhorse display for rendering surfaces [54]. Figure 4.50 is a contour plot of the loess fit to the galaxy velocities. The levels of the black contours range from 1440 km/sec to 1760 km/sec in steps of 20 km/sec; for the black and gray contours combined, the step size is 5 km/sec and the range is 1435 km/sec to 1760 km/sec. The two curve types enhance the perception of properties of the surface. The complete collection of curves conveys local properties, and the black curves by themselves convey global properties.

The contouring in Figure 4.50 is based on an evaluation of the surface on a grid whose east-west coordinates are 101 equally spaced values from $-25''$ to $25''$ and whose south-north coordinates are 181 equally spaced values from $-45''$ to $45''$. Thus there are $101 \times 181 = 18281$ grid points that move in steps of $0.5''$ along each coordinate axis. The grid lies slightly inside the rectangle that just contains the measurement locations of the data; such cropping for displaying a fit to data is discussed in Chapter 5.

The contours in Figure 4.50 are roughly symmetric about a line that runs from the region in the northeast where the velocities are lowest to the region in the southwest where the velocities are highest. From north to south along this line, the intersecting contours increase monotonically. The contours for the lowest velocities have a convex upward shape; as velocity increases, the convexity decreases until, for velocities around 1580 km/sec the contours are nearly linear. Then, as velocity increases further, the contours turn convex downward.

Suppose the only motions of NGC 7531 relative to the earth were a rotation and a recession. Then the velocity surface would be linear, and the contours in Figure 4.50 would be linear, parallel, equally spaced, and perpendicular to the projection of the axis of rotation onto our viewing plane. The actual contours bend and are not equally spaced. But in a small region centered near $(0'', 0'')$, with a diameter of about $10''$, they do appear to be linear, parallel, and equally spaced. This suggests that at the core of the galaxy the dominant motion is a rotation.



4.50 A contour plot displays the robust loess fit to the galaxy velocities. The aspect ratio has been chosen to make the number of data units per cm the same on both scales.